Title: EpiSense – Interactive Epidemiology and Biostatistics Software Suite
Programming Languages: HTML, CSS, JavaScript
Year of Creation: 2026

[Begin Source Code]

===== data_collection.html =====

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Data Collection & Geo Spatial Mapping - EpiSense</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Outfit:wght@300;400;600;700&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="main.css">
    <!-- Leaflet CSS -->
    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"
        integrity="sha256-p4NxAoJBhIIN+hmNHrzRCf9tD/miZyoHS5obTRR9BMY=" crossorigin="" />
    <script
src="https://cdnjs.cloudflare.com/ajax/libs/html2canvas/1.4.1/html2canvas.min.js">
</script>
    <style>
        :root {
            --step-active: var(--primary-blue);
            --step-inactive: rgba(255, 255, 255, 0.1);
        }

        .module-container {
            max-width: 1200px;
            margin: 2rem auto;
            padding: 2rem;
        }

        .gis-grid {
            display: grid;
            grid-template-columns: 350px 1fr;
            gap: 1.5rem;
            margin-top: 2rem;
        }

        #map {
            height: 650px;
            border-radius: 24px;
            border: 1px solid var(--glass-border);
            z-index: 1;
            background: #1a1a1a;
        }

        .sidebar {
            display: flex;
            flex-direction: column;
            gap: 1.5rem;
        }

        .control-card {
```

```
    background: var(--glass-bg);
    backdrop-filter: blur(12px);
    border: 1px solid var(--glass-border);
    border-radius: 20px;
    padding: 1.5rem;
}

.btn-gis {
    width: 100%;
    padding: 0.8rem;
    border-radius: 10px;
    border: none;
    background: var(--primary-pink);
    color: white;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.3s;
    margin-bottom: 0.5rem;
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 0.5rem;
}

.btn-gis:hover {
    transform: translateY(-2px);
    filter: brightness(1.1);
    box-shadow: 0 4px 15px rgba(244, 114, 182, 0.3);
}

.btn-gis.secondary {
    background: rgba(255, 255, 255, 0.05);
    border: 1px solid var(--glass-border);
}

.stat-item {
    display: flex;
    justify-content: space-between;
    margin-bottom: 0.5rem;
    font-size: 0.9rem;
    color: var(--text-dim);
}

.stat-value {
    color: var(--primary-blue);
    font-weight: 600;
}


.case-marker {
    width: 12px;
    height: 12px;
    border-radius: 50%;
    border: 2px solid white;
}

.dog-exposure {
    background: #f472b6;
}

.cat-exposure {
    background: #60a5fa;
}

.other-exposure {
```

```
    background: #fbbf24;
}

.legend-item {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    margin-bottom: 0.5rem;
    font-size: 0.85rem;
}

.dot {
    width: 10px;
    height: 10px;
    border-radius: 50%;
}

.back-btn {
    display: inline-flex;
    align-items: center;
    gap: 0.5rem;
    color: var(--text-dim);
    text-decoration: none;
    margin-bottom: 1rem;
    font-weight: 600;
}

#data-table-container {
    max-height: 200px;
    overflow-y: auto;
    margin-top: 1rem;
    border-radius: 10px;
    background: rgba(0, 0, 0, 0.2);
    font-size: 0.8rem;
}

.mini-table {
    width: 100%;
    border-collapse: collapse;
}

.mini-table th {
    position: sticky;
    top: 0;
    background: rgba(0, 0, 0, 0.5);
    text-align: left;
    padding: 5px;
}

.mini-table td {
    padding: 5px;
    border-bottom: 1px solid rgba(255, 255, 255, 0.05);
}

.info-pill {
    background: rgba(96, 165, 250, 0.1);
    color: #60a5fa;
    padding: 2px 8px;
    border-radius: 12px;
    font-size: 0.75rem;
    border: 1px solid rgba(96, 165, 250, 0.2);
}

.gis-select {
    width: 100%;
```

```
            padding: 0.8rem;
            border-radius: 10px;
            background: rgba(0, 0, 0, 0.3);
            border: 1px solid var(--glass-border);
            color: white;
            margin-bottom: 0.5rem;
            outline: none;
        }

        /* Mapping Modal */
        #mapping-modal {
            display: none;
            position: fixed;
            top: 0;
            left: 0;
            width: 100%;
            height: 100%;
            background: rgba(0, 0, 0, 0.8);
            z-index: 9999;
            backdrop-filter: blur(8px);
            align-items: center;
            justify-content: center;
        }

        .mapping-card {
            background: #1a1a1a;
            border: 1px solid var(--glass-border);
            border-radius: 24px;
            padding: 2rem;
            max-width: 500px;
            width: 90%;
        }

        .map-field {
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 1rem;
            align-items: center;
            margin-bottom: 1rem;
        }

        @media (max-width: 768px) {
            .module-container {
                padding: 1rem;
            }

            .gis-grid {
                grid-template-columns: 1fr;
            }

            #map {
                height: 400px;
                order: -1;
            }

            .sidebar {
                order: 1;
            }

            header h1 {
                font-size: 2rem !important;
            }
        }
    </style>
</head>
```

```
<body>
    <div class="background-blobs">
        <div class="blob blob-blue"></div>
        <div class="blob blob-pink"></div>
    </div>

    <div class="module-container">
        <a href="index.html" class="back-btn">
            <svg width="20" height="20" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2">
                <path d="M19 12H5"></path>
                <polyline points="12 19 5 12 12 5"></polyline>
            </svg>
            Back to Dashboard
        </a>

        <header
            style="margin-bottom: 2rem; display: flex; flex-direction: column; align-
items: center; text-align: center;">
            <h1
                style="font-size: 2.5rem; background: linear-gradient(to right, #60a5fa,
#f472b6); -webkit-background-clip: text; background-clip: text; -webkit-text-fill-color:
transparent;">
                Data Collection & Geo Spatial Mapping
            </h1>
            <p class="subtitle" style="margin: 0 auto; max-width: 800px;">
                Understanding geographic outbreaks through case clustering and hotspot
detection.
                Enter case reports or use dummy data to visualize the spread of public
health events.
            </p>
        </header>

        <div class="gis-grid">
            <div class="sidebar">
                <!-- Data Entry -->
                <div class="control-card">
                    <h3>Spatial Data Entry</h3>
                    <p style="font-size: 0.85rem; color: var(--text-dim); margin-bottom:
1rem;">
                        Simulate an outbreak or upload incident reports.
                    </p>

                    <button class="btn-gis" onclick="generateDummyData()">
                        <svg width="18" height="18" viewBox="0 0 24 24" fill="none"
stroke="currentColor"
                            stroke-width="2">
                            <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"></path>
                            <polyline points="7 10 12 15 17 10"></polyline>
                            <line x1="12" y1="15" x2="12" y2="3"></line>
                        </svg>
                        Generate Dummy Outbreak
                    </button>

                    <div style="display: flex; gap: 0.5rem; margin-bottom: 1rem;">
                        <input type="file" id="csv-upload" hidden accept=".csv">
                        <button class="btn-gis secondary"
onclick="document.getElementById('csv-upload').click()"
                            style="flex: 1;">
                            Upload CSV
                        </button>
                    </div>

                    <button class="btn-gis secondary" onclick="downloadSampleCSV()"
```

```
                        style="margin-bottom: 0.5rem; font-size: 0.8rem; border-style:
dashed; border-color: var(--primary-blue);">
                        <svg width="14" height="14" viewBox="0 0 24 24" fill="none"
stroke="currentColor"
                            stroke-width="2">
                            <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"></path>
                            <polyline points="7 10 12 15 17 10"></polyline>
                            <line x1="12" y1="15" x2="12" y2="3"></line>
                        </svg>
                        Download Sample CSV Template
                    </button>

                    <button class="btn-gis secondary" onclick="openManualEntry()"
                        style="margin-bottom: 1.5rem; font-size: 0.8rem; border-style:
solid; border-color: var(--primary-pink); color: var(--primary-pink);">
                        <svg width="14" height="14" viewBox="0 0 24 24" fill="none"
stroke="currentColor"
                            stroke-width="2" stroke-linecap="round" stroke-
linejoin="round">
                            <path d="M12 5v14M5 12h14" />
                        </svg>
                        Manual Data Entry
                    </button>

                    <h3>Focus Region</h3>
                    <select id="state-selector" class="gis-select"
onchange="focusState(this.value)">
                        <option value="">Zoom to Entire India</option>
                    </select>

                    <div id="data-table-container">
                        <table class="mini-table" id="case-table">
                            <thead>
                                <tr>
                                    <th>ID</th>
                                    <th>Type</th>
                                    <th>Lat</th>
                                    <th>Long</th>
                                    <th>State</th>
                                </tr>
                            </thead>
                            <tbody id="table-body">
                                <tr>
                                    <td colspan="5" style="text-align: center; color:
var(--text-dim);">No data loaded
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>

                <!-- Clustering Controls -->
                <div class="control-card">
                    <h3>Clustering Algorithm</h3>
                    <div style="margin-bottom: 1rem;">
                        <label style="font-size: 0.85rem;">Distance Threshold</label>
                        <input type="range" id="cluster-dist" min="5" max="5000"
value="1000" style="width: 100%;">
                        <div class="stat-item"><span id="dist-val">1.00 km</span></div>
                    </div>

                    <div style="margin-bottom: 1rem; display: flex; align-items: center;
gap: 0.5rem;">
                        <input type="checkbox" id="cluster-by-type" style="width: 16px;
```

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height: 16px; cursor: pointer;">
                        <label for="cluster-by-type"
                            style="font-size: 0.85rem; cursor: pointer; color: var(--text-
dim);">Cluster by Exposure
                            Type</label>
                    </div>

                    <button class="btn-gis" onclick="runClustering()" id="cluster-btn"
disabled>
                        Detect Clusters
                    </button>

                    <div id="clustering-stats" style="margin-top: 1rem; display: none;">
                        <div class="stat-item">Total Cases: <span class="stat-value"
id="stat-total">0</span></div>
                        <div class="stat-item">Clusters Found: <span class="stat-value"
id="stat-clusters">0</span>
                        </div>
                        <div class="stat-item">Hotspot Cases: <span class="stat-value"
id="stat-hotspot">0</span></div>
                    </div>
                </div>

                <div class="control-card">
                    <h3>Legend</h3>
                    <div id="dynamic-legend">
                        <p style="font-size: 0.8rem; color: var(--text-dim);">No data
loaded to show legend.</p>
                    </div>
                </div>

                <button class="btn-gis secondary" onclick="resetData()"
                    style="margin-bottom: 1rem; color: #ef4444; border-color: rgba(239,
68, 68, 0.2);">
                    <svg width="18" height="18" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2">
                        <path d="M3 6h18"></path>
                        <path d="M19 6v14c0 1-1 2-2 2H7c-1 0-2-1-2-2V6"></path>
                        <path d="M8 6V4c0-1 1-2 2-2h4c1 0 2 1 2 2v2"></path>
                    </svg>
                    Reset Map & Data
                </button>

                <button class="btn-gis secondary" onclick="downloadCurrentDataCSV()"
                    style="margin-bottom: 0.5rem; border-color: #10b981; color: #10b981;">
                    <svg width="18" height="18" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2">
                        <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"></path>
                        <polyline points="7 10 12 15 17 10"></polyline>
                        <line x1="12" y1="15" x2="12" y2="3"></line>
                    </svg>
                    Download Current Data (CSV)
                </button>

                <button class="btn-gis" onclick="downloadMapPNG()" style="background:
var(--primary-blue);">
                    <svg width="18" height="18" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2">
                        <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"></path>
                        <polyline points="7 10 12 15 17 10"></polyline>
                        <line x1="12" y1="15" x2="12" y2="3"></line>
                    </svg>
                    Download Map (PNG)
                </button>
```

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                <!-- Educational Section (Moved to Sidebar) REMOVED -->
            </div>

            <div style="position: relative; display: flex; flex-direction: column; gap:
1.5rem;">
                <div id="map"></div>
                <div id="map-loader"
                    style="display:none; position: absolute; top:50%; left:50%;
transform:translate(-50%, -50%); z-index:1000; background: rgba(0,0,0,0.8); padding: 20px;
border-radius: 12px; color: white;">
                    Loading India Geometry...
                </div>

                <!-- Educational Section (Consolidated below Map) -->
                <div class="control-card" style="padding: 1.5rem; border-color: rgba(96,
165, 250, 0.3); width: 100%;">
                    <h3
                        style="color: var(--primary-blue); margin-bottom: 0.8rem; font-
size: 1.2rem; border-bottom: 1px solid var(--glass-border); padding-bottom: 0.5rem;
display: flex; align-items: center; gap: 0.5rem;">
                        <svg width="20" height="20" viewBox="0 0 24 24" fill="none"
stroke="currentColor"
                            stroke-width="2">
                            <path d="M12 22s8-4 8-10V5l-8-3-8 3v7c0 6 8 10 8 10z"></path>
                        </svg>
                        Algorithm Insights & Spatial Logic
                    </h3>
                    <div style="display: grid; grid-template-columns: 1fr 1.2fr; gap:
2rem;">
                        <div style="font-size: 0.9rem; color: var(--text-dim); line-
height: 1.6;">
                            <p style="margin-bottom: 1rem;">
                                This module utilizes a <strong>Proximity-Linked
Aggregation</strong> (DBSCAN logic) to
                                identify high-density geographic clusters.
                            </p>
                            <div
                                style="background: rgba(255,255,255,0.03); padding: 1rem;
border-radius: 12px; border: 1px solid rgba(255,255,255,0.05);">
                                <ul style="list-style: none; padding: 0;">
                                    <li style="margin-bottom: 0.8rem;">📍 <strong>Spatial
Join:</strong> Pairs points
                                        within threshold distance.</li>
                                    <li style="margin-bottom: 0.8rem;">📏
<strong>Threshold:</strong> Clusters only form
                                        if points overlap in chosen radius.</li>
                                    <li>🔥 <strong>Hotspot:</strong> Confirmed when
density reaches the threshold (min.
                                        3 points).</li>
                                </ul>
                            </div>
                        </div>
                        <div
                            style="padding: 1.2rem; background: rgba(244, 114, 182, 0.05);
border-left: 4px solid var(--primary-pink); border-radius: 10px;">
                            <span
                                style="font-size: 0.9rem; font-weight: 700; color: var(--
primary-pink); display: block; margin-bottom: 0.5rem;">EPIDEMIOLOGICAL
                                VALUE:</span>
                            <p style="font-size: 0.9rem; color: var(--text-dim); line-
height: 1.6;">
                                Spatial clustering is vital for identifying <strong>Shared
Exposure Points</strong>. By
                                pinpointing where incidents occur in close temporal and
spatial proximity, public health
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                                teams can deploy targeted interventions, such as vector
control or sanitation
                                improvements, effectively breaking the chain of
transmission.
                            </p>
                            <p
                                style="font-size: 0.85rem; color: var(--text-dim); margin-
top: 0.5rem; font-style: italic;">
                                Note: Adjust the <strong>Distance Threshold</strong> to
vary analysis between
                                street-level (e.g. 100m) and city-wide (e.g. 5km) trends.
                            </p>
                        </div>
                    </div>
                </div>
            </div>
        </div>

        <!-- Mapping Modal -->
        <div id="mapping-modal">
            <div class="mapping-card">
                <h2>Field Mapping</h2>
                <p style="color: var(--text-dim); margin-bottom: 1.5rem;">Select
corresponding columns from your CSV.
                </p>

                <div id="mapping-fields">
                    <div class="map-field">
                        <label>Case ID / Serial</label>
                        <select id="map-id" class="gis-select"></select>
                    </div>
                    <div class="map-field">
                        <label>Latitude <span style="color:red">*</span></label>
                        <select id="map-lat" class="gis-select"></select>
                    </div>
                    <div class="map-field">
                        <label>Longitude <span style="color:red">*</span></label>
                        <select id="map-lng" class="gis-select"></select>
                    </div>
                    <div class="map-field">
                        <label>Exposure Type</label>
                        <select id="map-type" class="gis-select"></select>
                    </div>
                    <div class="map-field">
                        <label>State Name</label>
                        <select id="map-state" class="gis-select"></select>
                    </div>
                </div>

                <div style="display: flex; gap: 1rem; margin-top: 1rem;">
                    <button class="btn-gis secondary"
                        onclick="document.getElementById('mapping-
modal').style.display='none'">Cancel</button>
                    <button class="btn-gis" style="margin-bottom:0"
onclick="applyMapping()">Load Mapped Data</button>
                </div>
            </div>
        </div>

        <!-- Manual Entry Modal -->
        <div id="manual-entry-modal"
            style="display: none; position: fixed; top: 0; left: 0; width: 100%; height:
100%; background: rgba(0,0,0,0.8); z-index: 5000; align-items: center; justify-content:
center;">
            <div class="mapping-card">
```

```
                <h2>Manual Data Entry</h2>
                <p style="color: var(--text-dim); margin-bottom: 1.5rem;">Enter a single
case report manually or capture
                    your current location.</p>

                <div id="manual-fields">
                    <div class="map-field">
                        <label>Case ID / Serial</label>
                        <input type="text" id="manual-id" class="gis-select"
placeholder="e.g. ID-101 (Optional)"
                            style="background: rgba(255,255,255,0.05); color: white;
border: 1px solid var(--glass-border); border-radius: 8px; padding: 0.8rem;">
                    </div>

                    <button class="btn-gis" onclick="captureLocation()"
                        style="background: #10b981; font-size: 0.85rem; padding: 0.5rem;
margin-bottom: 1rem;">
                        <svg width="14" height="14" viewBox="0 0 24 24" fill="none"
stroke="currentColor"
                            stroke-width="2">
                            <circle cx="12" cy="12" r="10"></circle>
                            <circle cx="12" cy="12" r="3"></circle>
                        </svg>
                        Capture Current Device Location
                    </button>

                    <div style="display: grid; grid-template-columns: 1fr 1fr; gap:
1rem;">
                        <div class="map-field">
                            <label>Latitude <span style="color:red">*</span></label>
                            <input type="number" id="manual-lat" class="gis-select"
step="any"
                                placeholder="e.g. 22.5726"
                                style="background: rgba(255,255,255,0.05); color: white;
border: 1px solid var(--glass-border); border-radius: 8px; padding: 0.8rem;">
                        </div>
                        <div class="map-field">
                            <label>Longitude <span style="color:red">*</span></label>
                            <input type="number" id="manual-lng" class="gis-select"
step="any"
                                placeholder="e.g. 88.3639"
                                style="background: rgba(255,255,255,0.05); color: white;
border: 1px solid var(--glass-border); border-radius: 8px; padding: 0.8rem;">
                        </div>
                    </div>

                    <div class="map-field">
                        <label>Exposure Type <span style="color:red">*</span></label>
                        <select id="manual-type" class="gis-select">
                            <option value="Dog">Dog</option>
                            <option value="Cat">Cat</option>
                            <option value="Monkey">Monkey</option>
                            <option value="Fox">Fox</option>
                            <option value="Other">Other</option>
                        </select>
                    </div>
                    <div class="map-field">
                        <label>State Name</label>
                        <input type="text" id="manual-state" class="gis-select"
placeholder="e.g. WEST BENGAL"
                            style="background: rgba(255,255,255,0.05); color: white;
border: 1px solid var(--glass-border); border-radius: 8px; padding: 0.8rem;">
                    </div>
                </div>
```

```
    box-shadow: 0 8px 32px 0 rgba(31, 38, 135, 0.1);
    display: flex;
    flex-direction: column;
    gap: 2rem;
}

.visualization-container {
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 2rem;
    padding-bottom: 2rem;
    border-bottom: 1px solid rgba(0, 0, 0, 0.05);
}

.metric {
    flex: 1;
    text-align: center;
    background: rgba(255, 255, 255, 0.4);
    /* Glassy child */
    backdrop-filter: blur(4px);
    padding: 2rem;
    border-radius: 16px;
    border: 1px solid rgba(255, 255, 255, 0.5);
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.02);
}

.metric .label {
    display: block;
    font-size: 1rem;
    text-transform: uppercase;
    letter-spacing: 1px;
    color: var(--secondary-color);
    margin-bottom: 0.5rem;
    font-weight: 600;
}

.metric .value {
    display: block;
    font-size: 5rem;
    font-weight: 700;
    line-height: 1;
    color: var(--primary-color);
    text-shadow: 2px 2px 0px rgba(0, 0, 0, 0.05);
}

.metric .formula-display {
    display: block;
    margin-top: 1rem;
    font-family: 'Courier New', monospace;
    color: var(--text-muted);
    font-size: 1.2rem;
    background: rgba(0, 0, 0, 0.03);
    padding: 0.5rem;
    border-radius: 8px;
}

.metric .dynamic-params {
    display: block;
    margin-top: 0.5rem;
    font-family: 'Courier New', monospace;
    color: var(--primary-color);
    font-weight: 600;
    font-size: 1rem;
    background: rgba(0, 123, 255, 0.1);
```

```
    padding: 0.5rem;
    border-radius: 8px;
    letter-spacing: 0.5px;
}

.visual-representation {
    flex: 1;
    height: 280px;
    position: relative;
    display: flex;
    flex-direction: column;
    justify-content: flex-start;
    /* changed from center to stack from top */
    align-items: center;
    border-radius: 16px;
    background: rgba(255, 255, 255, 0.5);
    box-shadow: inset 0 2px 10px rgba(0, 0, 0, 0.03);
    border: 1px solid rgba(255, 255, 255, 0.5);
    padding: 10px;
    overflow: hidden;
    /* Prevent spill */
}

.visual-legend {
    width: 100%;
    text-align: center;
    color: #777;
    font-size: 14px;
    font-style: italic;
    margin-top: 10px;
    flex-shrink: 0;
    /* Don't shrink the legend */
}

#sample-canvas {
    width: 100%;
    flex: 1;
    /* Take remaining height */
    border-radius: 12px;
}

.controls-container {
    display: flex;
    flex-direction: column;
    gap: 2rem;
}

.control-group {
    display: flex;
    flex-direction: column;
    gap: 1rem;
    background: rgba(255, 255, 255, 0.5);
    /* Semi-transparent */
    backdrop-filter: blur(4px);
    padding: 1.5rem;
    border-radius: 12px;
    border: 1px solid rgba(255, 255, 255, 0.6);
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.03);
}

.control-group label {
    display: flex;
    justify-content: space-between;
    font-weight: 600;
    font-size: 1.1rem;
```

```
    color: var(--text-main);
}

.control-group label span {
    color: var(--primary-color);
    background: rgba(0, 123, 255, 0.1);
    padding: 0.2rem 0.6rem;
    border-radius: 6px;
}

.description-text {
    font-size: 0.9rem;
    color: var(--text-muted);
    margin-top: -0.5rem;
    margin-bottom: 0.5rem;
    line-height: 1.4;
}

/* Custom Range Slider - High Visibility & Refined */
input[type=range] {
    -webkit-appearance: none;
    appearance: none;
    width: 100%;
    background: transparent;
    margin: 10px 0;
    height: 24px;
    /* Adjusted container height */
}

input[type=range]:focus {
    outline: none;
}

/* Track Styling */
input[type=range]::-webkit-slider-runnable-track {
    width: 100%;
    height: 8px;
    /* Thinner track */
    cursor: pointer;
    background: #bdc3c7;
    border-radius: 4px;
    border: 1px solid #95a5a6;
    box-shadow: inset 0 1px 3px rgba(0, 0, 0, 0.1);
}

/* Firefox Track */
input[type=range]::-moz-range-track {
    width: 100%;
    height: 8px;
    /* Thinner track */
    cursor: pointer;
    background: #bdc3c7;
    border-radius: 4px;
    border: 1px solid #95a5a6;
    box-shadow: inset 0 1px 3px rgba(0, 0, 0, 0.1);
}

/* Thumb Styling */
input[type=range]::-webkit-slider-thumb {
    height: 22px;
    /* Smaller thumb */
    width: 22px;
    border-radius: 50%;
    background: var(--primary-color);
    cursor: pointer;
```

```
    -webkit-appearance: none;
    appearance: none;
    margin-top: -8px;
    /* Centered: (8px track - 22px thumb) / 2 = -7px. -8px for visual balance with border.
*/
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.2);
    border: 3px solid #ffffff;
    /* Slightly thinner border */
    transition: transform 0.1s ease, background 0.2s, box-shadow 0.2s;
}

input[type=range]::-moz-range-thumb {
    height: 22px;
    width: 22px;
    border: 3px solid #ffffff;
    border-radius: 50%;
    background: var(--primary-color);
    cursor: pointer;
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.2);
}

input[type=range]::-webkit-slider-thumb:hover {
    transform: scale(1.15);
    background: var(--secondary-color);
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.3);
}

/* Specific colors for different sliders (optional but good for visibility) */
#prevalence::-webkit-slider-thumb {
    background: var(--primary-color);
}

#prevalence::-webkit-slider-thumb:hover {
    background: #0056b3;
}

#precision::-webkit-slider-thumb {
    background: var(--secondary-color);
    box-shadow: 0 4px 8px rgba(232, 62, 140, 0.5);
}

#precision::-webkit-slider-thumb:hover {
    background: #c91f6e;
}

/* Colored tracks for active feel?
   Let's keep track neutral to let thumb pop, or add a fill?
   Webkit doesn't support fill-lower easy without JS or complex CSS.
   Distinct grey is "clearly visible".
*/

.info-section {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 1.5rem;
    margin-top: 1rem;
    padding-top: 2rem;
    border-top: 1px solid rgba(0, 0, 0, 0.05);
}

.info-card {
    background: rgba(255, 255, 255, 0.6);
    padding: 1.5rem;
    border-radius: 12px;
    border: 1px solid rgba(0, 0, 0, 0.05);
```

```
}

.info-card h3 {
    margin: 0 0 0.5rem 0;
    color: var(--primary-color);
    font-size: 1.1rem;
}

.info-card p {
    margin: 0;
    font-size: 0.9rem;
    color: var(--text-muted);
    line-height: 1.5;
}

footer {
    margin-top: 2rem;
    font-size: 0.9rem;
    color: var(--text-muted);
    text-align: center;
    padding: 1rem;
    background: rgba(255, 255, 255, 0.8);
    border-radius: 50px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.05);
}

.developer-name {
    color: var(--secondary-color);
    font-weight: 700;
}

@media (max-width: 768px) {
    #app {
        padding: 1rem;
        gap: 1.5rem;
    }

    h1 {
        font-size: 1.8rem;
        line-height: 1.2;
    }

    .subtitle {
        font-size: 1rem;
    }

    .glass-panel {
        padding: 1.5rem;
        border-radius: 20px;
    }

    /* Stack Visualization on Mobile */
    .visualization-container {
        flex-direction: column;
        gap: 1.5rem;
        padding-bottom: 1.5rem;
    }

    .metric {
        width: 100%;
        padding: 1.5rem;
        box-sizing: border-box;
    }

    .metric .value {
```

```
        font-size: 3.5rem;
        /* Smaller for mobile to prevent wrapping */
    }

    .metric .formula-display {
        font-size: 0.9rem;
        padding: 0.4rem;
        overflow-x: auto;
        /* Allow formula scroll if too long */
    }

    .metric .dynamic-params {
        font-size: 0.8rem;
        padding: 0.4rem;
        white-space: normal;
        word-break: break-word;
    }

    .visual-representation {
        width: 100%;
        height: 300px;
        /* Taller on mobile for better visibility */
        margin-top: 1rem;
    }

    .study-selector-container {
        flex-direction: column;
        gap: 0.5rem;
        width: 100%;
        box-sizing: border-box;
    }

    .study-dropdown {
        width: 100%;
        text-align: center;
        font-size: 1.1rem;
        /* Larger touch target */
        padding: 0.8rem;
    }

    /* Bigger touch targets for sliders */
    input[type=range] {
        height: 30px;
        margin: 15px 0;
    }

    input[type=range]::-webkit-slider-thumb {
        width: 28px;
        height: 28px;
    }

    .control-group label {
        font-size: 1rem;
    }

    .info-section {
        grid-template-columns: 1fr;
        /* Stack info cards */
    }

    footer {
        padding: 1.5rem;
        font-size: 0.85rem;
    }
}
```

```
/* New Styles for Study Selector */
.study-selector-container {
    margin-top: 1.5rem;
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 1rem;
    background: rgba(255, 255, 255, 0.4);
    padding: 0.5rem 1rem;
    border-radius: 50px;
    border: 1px solid rgba(255, 255, 255, 0.6);
}

.selector-label {
    font-weight: 600;
    color: var(--text-main);
}

.study-dropdown {
    padding: 0.5rem 1rem;
    border-radius: 8px;
    border: 1px solid #bdc3c7;
    background: rgba(255, 255, 255, 0.9);
    font-family: var(--font-family);
    font-size: 1rem;
    color: var(--text-main);
    cursor: pointer;
    outline: none;
    transition: all 0.2s ease;
}

.study-dropdown:focus {
    border-color: var(--primary-color);
    box-shadow: 0 0 0 3px rgba(0, 123, 255, 0.1);
}

/* Checkbox / Toggle Switch Styles */
.checkbox-wrapper {
    display: flex;
    align-items: center;
    /* or flex-start if label is multiline */
    gap: 0.8rem;
    margin-bottom: 0.5rem;
    justify-content: space-between;
    /* Put toggle on right? or keep left? Standard is usually left or right. User didn't
specify, but preventing shrink is key. */
}

.tgl {
    display: none;
}

.tgl,
.tgl:after,
.tgl:before,
.tgl *,
.tgl *:after,
.tgl *:before,
.tgl+.tgl-btn {
    box-sizing: border-box;
}

.tgl::-moz-selection,
.tgl:after::-moz-selection,
```

```
.tgl:before::-moz-selection,
.tgl *::-moz-selection,
.tgl *:after::-moz-selection,
.tgl *:before::-moz-selection,
.tgl+.tgl-btn::-moz-selection {
    background: none;
}

.tgl::selection,
.tgl:after::selection,
.tgl:before::selection,
.tgl *::selection,
.tgl *:after::selection,
.tgl *:before::selection,
.tgl+.tgl-btn::selection {
    background: none;
}

.tgl+.tgl-btn {
    outline: 0;
    display: block;
    width: 44px;
    height: 24px;
    position: relative;
    cursor: pointer;
    user-select: none;
    flex-shrink: 0;
    /* Critical: Prevent compression */
}

.tgl+.tgl-btn:after,
.tgl+.tgl-btn:before {
    position: relative;
    display: block;
    content: "";
    width: 50%;
    height: 100%;
}

.tgl+.tgl-btn:after {
    left: 0;
}

.tgl+.tgl-btn:before {
    display: none;
}

.tgl:checked+.tgl-btn:after {
    left: 50%;
}

.tgl-ios+.tgl-btn {
    background: #fbfbfb;
    border-radius: 2em;
    padding: 2px;
    transition: all .4s ease;
    border: 1px solid #e8eae9;
}

.tgl-ios+.tgl-btn:after {
    border-radius: 2em;
    background: #fbfbfb;
    transition: left 0.3s cubic-bezier(0.175, 0.885, 0.320, 1.275), padding 0.3s ease,
margin 0.3s ease;
    box-shadow: 0 0 0 1px rgba(0, 0, 0, 0.1), 0 4px 0 rgba(0, 0, 0, 0.08);
```

```
}

.tgl-ios+.tgl-btn:active {
    box-shadow: inset 0 0 0 2em #e8eae9;
}

.tgl-ios+.tgl-btn:active:after {
    padding-right: .8em;
}

.tgl-ios:checked+.tgl-btn {
    background: var(--primary-color);
}

.tgl-ios:checked+.tgl-btn:active {
    box-shadow: none;
}

.tgl-ios:checked+.tgl-btn:active:after {
    margin-left: -.8em;
}

/* Number Input Styling */
input[type="number"] {
    width: 100%;
    padding: 0.6rem 0.8rem;
    border-radius: 8px;
    border: 1px solid #ccc;
    font-family: var(--font-family);
    font-size: 1rem;
    background: rgba(255, 255, 255, 0.8);
    margin-top: 0.5rem;
}

input[type="number"]:focus {
    border-color: var(--primary-color);
    outline: none;
    box-shadow: 0 0 0 3px rgba(0, 123, 255, 0.1);
}

/* Footer */
footer {
    text-align: center;
    padding: 2rem;
    margin-top: 2rem;
    font-size: 0.95rem;
    background: rgba(255, 255, 255, 0.4);
    border-top: 1px solid rgba(255, 255, 255, 0.5);
    width: 100%;
}

.footer-line {
    margin-bottom: 0.4rem;
    font-weight: 600;
    letter-spacing: 0.5px;
}

.blue-text {
    color: var(--primary-color);
    font-size: 1.1rem;
    /* Slightly larger for 'Developed By' */
}

.pink-text {
    color: var(--secondary-color);
```

```
}

.disclaimer {
    margin-top: 1.5rem;
    padding-top: 1rem;
    border-top: 1px solid rgba(0, 0, 0, 0.05);
    font-size: 0.85rem;
    color: #666;
    max-width: 800px;
    margin-left: auto;
    margin-right: auto;
    line-height: 1.4;
}

.disclaimer a {
    color: var(--primary-color);
    text-decoration: none;
    font-weight: 600;
}

.disclaimer a:hover {
    text-decoration: underline;
}
----------------------------------------
[End of Source Code]
----------------------------------------
```